04/18/2022

Monday, April 18, 2022 10:20 AM

M168_Notes_8.4

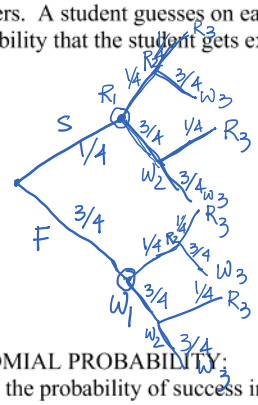
8.4 The Binomial Distribution

In a Bernoulli trial we have the following:

- The same experiment repeated several times. ✓
- The only possible outcomes of these experiments are success or failure. ✓
- The repeated trials are independent so the probability of success remains the same for each trial. ✓

*Example*

A multiple choice test has 3 questions, each with 4 possible answers. A student guesses on each question. What is the probability that the student gets exactly two questions correct?




$$P(R_1 \cap R_2 \cap W_3) + P(R_1 \cap W_2 \cap R_3) + P(W_1 \cap R_2 \cap R_3)$$

$$\frac{1}{4} \cdot \frac{1}{4} \cdot \frac{3}{4} \qquad \frac{1}{4} \cdot \frac{3}{4} \cdot \frac{1}{4} \qquad \frac{3}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} = 3 \cdot \left(\frac{1}{4} \cdot \frac{1}{4} \cdot \frac{3}{4}\right) = \frac{9}{64}$$

BINOMIAL PROBABILITY:

If p is the probability of success in a single trial of a binomial (Bernoulli) experiment, the probability of x successes and n-x failures in n independent repeated trials of the same experiment is

$$\boxed{C(n,x)}(p)^x(1-p)^{n-x} = \text{binompdf}(n,p,x)$$

$p = 1/4$, $n = 3$, $x = 2$

$$\text{binompdf}(3, 1/4, 2) = 3 \cdot (1/4)^2 \cdot \left(\frac{3}{4}\right) = \frac{9}{64}$$

*Example*

The first half of July was very dry in college station. If each day there was a 20% chance of rain, what is the probability of no rain in the first 15 days in July?

DEFINE SUCCESS:

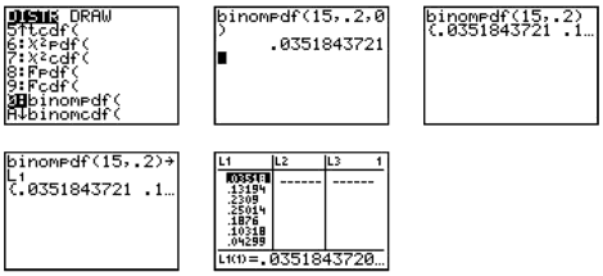
n = number of trials = 15

x = number of successes = 15

p = probability of success = 0.8

Ans = 0.0351.

binompdf(n, p, x) on the calculator




n = 15

x = 0

p = .2

Ans = 0.0351

What is the probability of at most 2 rain days?

x = number of successes = 0, 1, 2

$L_1(1) + L_1(2) + L_1(3) = 0.3980\ldots$

$P(x=0) \quad P(x=1) \quad P(x=2)$

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L1(1)+L1(2)+L1(3
)
     .3980232093
```

binomcdf(n, p, x) will give you the sum of the probabilities from 0 to x.

```
binomcdf(15,.2,2
)
     .3980232116
```

For a binomial probability question you must do the following:

- Decide that it is a Bernoulli trial.
- Define what success is.
- Find the number of times the experiment is done, n.
- Find the probability of success, p.
- Determine the number of successes you need to find, x.

*Example*

A new drug being tested causes a serious side effect in 5% of patients. What is the probability that in a sample of 10 patients none get the side effect from taking the drug?

define success = no side effect,

n = number of trials = 10

x = number of successes = 10

p = probability of success = 0.95

binompdf (10, 0.95, 10) = 0.5987

define success = side effect,

n = number of trials = 10

x = number of successes = 0

p = probability of success = 0.05

binompdf (10, 0.05, 0) = 0.5987

$\mu = 0.5$

$\sigma = \sqrt{10 \times (0.05) \times 0.95}$

If X is a binomial random variable associated with a binomial experiment consisting of n trials with probability of success p and probability of failure q=1-p, then the mean (expected value) and standard deviation associated with the experiment are:

$E(X)$ $\boxed{\mu = np}, \sigma = \sqrt{(npq)}$

Example - let the random variable X be the number of girls in a 6 child family. Find the probability distribution table, probability histogram and the mean and standard deviation for the number of girls in the family.

$n = 6$

$\mu = 3$

$p = 0.5$